

Temperature scaling & Feynman–Kac correctors on CoBit (hard Sudoku)

One-page summary • branch `tasks/fkc-temperature` • raw (non-EMA) weights, $K=16$, 180 steps, hard difficulty

Setup

CoBit = continuous VE diffusion over analog bits; denoiser posterior D_θ , score $s_\theta = (D_\theta - x)/\sigma^2$; prompt clamped. Goal: principled low-temperature / annealed / guided sampling. We implemented **A1** (local score-level temperature $\kappa(\sigma) = \frac{v+\sigma^2}{\tau v+\sigma^2}$, $v=0.25$), a **Feynman–Kac Euler–Maruyama SMC sampler** (annealed p_c^β and CFG/Prop-3.1 geometric average $q_u^{1-w} q_c^w$; **em** and **edm_churn** proposals; weight $\frac{1}{2}\beta(\beta-1)\Delta\sigma^2\|s\|_{\text{free}}^2$), and an **entropy-band** resampling option. 16 correctness gates pass (incl. analytic 1-D Gaussian $\text{Var} \rightarrow (v + \sigma^2)/\beta$).

Headline results

config (hard, $K=16$)	maj@16	pass@16	distinct/16	note
$\beta=1$ voting (no reweighting)	90.8%	93.6%	5.45	reference
$\beta=1.02$, resampling OFF	90.6%	93.4%	5.44	ties voting; diversity kept
$\beta=1.02$, resample (ESS<K/2)	84.8%	87.5%	4.07	resampling erodes diversity
$\beta=1.05$, resample	71.5%	72.3%	2.70	collapse ($\rightarrow 2$ samples)
CFG-FKC $w=1.5$, resample	$\sim 64\%$	$\sim 64\%$	~ 2	pass=maj (full collapse)
A1 τ : 1.0 $\rightarrow$ 0.3 (deterministic)	exact 41.2 $\rightarrow$ 43.4%, valid	43.4 $\rightarrow$ 47.4%		monotone, no collapse

Main conclusions

1. **The culprit is resampling, not tempering.** Genealogical resampling collapses the 16-particle cloud from ~ 5.4 to ~ 2 distinct solutions, destroying the coverage that makes voting work. Particle *mean* accuracy stays $\sim 67\%$ throughout — particles don't get worse, they get *correlated*.
2. **Tempered-drift + voting (resampling off) ties $\beta=1$ voting** (90.6 vs 90.8, within $n=512$ noise), with diversity preserved. So mild tempering neither clearly helps nor hurts; the FK *correction* (resample-toward-the-weight) is the part that hurts.
3. **The FKC path weight is not a correctness signal.** On the resampling-off run (full, unreset cumulative potential): highest-weight particle $63.9\% < \text{mean } 66.9\%$; weight-weighted vote $88.5\% < \text{plain vote } 90.6\%$. Using the weight to select/vote *hurts*.
4. **Coverage is diversity-bound.** Even $\beta=1$ yields only 5.45 distinct of 16 (limited stochastic diversity), so raw K has diminishing returns ($K=32$ barely beat $K=16$).
5. **A1 is the one lever that helps** — small, monotone, collapse-free — because it is a gentle deterministic score-sharpening with no importance weights to degenerate.

What we do *not* claim (post-review)

Not established: that model confidence is unrelated to correctness (the voting gain suggests the opposite); nor that correctly-implemented FKC fails on CoBit. Untested links: **edm_churn** weight at $\beta>1$ / finite $N=180$ (asymptotic only); carry self-conditioning vs the Markov-score assumption; mode-reweighting (gate tested variance only). CFG results confounded by a weak 20k-step, dropout-trained checkpoint.

Recommended next steps

(i) **Verify@K** using the Sudoku rule-checker (a label-free, near-exact correctness oracle) $\Rightarrow$ realizes $\text{pass}@K \approx \mathbf{94\%}$ for free (Track C, Phase 0). (ii) **Reward-guided SMC** (SVDD): replace the likelihood weight with the constraint-satisfaction reward — steer trajectories toward *correct* modes, not confident ones. (iii) Decisive FKC audits: **never**-resample β -sweep (does any β beat voting?); toy-mixture mode-reweighting & N -convergence; self-conditioning ablation.

Full detail: `docs/fkc_temperature_report.md` (methods, math, all tables, threats to validity, reproduction). Commits `ead8f4b` $\rightarrow$ `aa87579`.